Automated IT Support Ticket Classification and Routing Using BERT and Workflow Automation

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***Abstract*—The automation of IT support ticket classification and routing plays a crucial role in improving service quality and reducing response time in large-scale organizations. This research presents a novel approach that combines transformer-based Natural Language Processing (NLP) with low-code workflow automation to enhance Information Technology Service Management (ITSM). The system leverages Bidirectional Encoder Representations from Transformers (BERT) for contextual understanding of support tickets, and integrates the trained model into an n8n-based automation pipeline. This enables automatic categorization and routing of incoming support tickets to the appropriate resolution teams. The proposed work extends previous research in AI-based ticket classification, such as AutoML-driven methods, by introducing semantic deep learning models and real-time workflow automation. The solution is designed to be scalable, interpretable, and practically deployable in enterprise IT environments.**

I. Introduction

IT Service Management (ITSM) processes rely heavily on efficient handling of support tickets to maintain uninterrupted services and customer satisfaction. A support ticket, generated when an end-user reports an issue or requests a change, forms the backbone of IT support operations. With the growing volume of tickets in modern enterprises, manual classification and routing have become increasingly challenging and time-consuming. Human-based triage not only increases operational costs but also introduces inconsistency and delay in service resolution.

Recent advancements in Artificial Intelligence (AI) have opened new pathways for automating ticket categorization tasks. The study by Truss and Böhm (2021) introduced an AI-based framework employing Google Vertex AutoML for customer support ticket classification. Their work demonstrated that AutoML can automate model training and optimization without significant human intervention. However, AutoML systems have inherent limitations in understanding contextual semantics within text and integrating directly with workflow tools.

To overcome these limitations, the present research explores the application of transformer-based models—specifically BERT—for deep contextual understanding of IT support tickets. The study also integrates this intelligent classifier within an n8n automation environment, enabling automatic ticket routing to the relevant support teams via real-time communication channels such as Telegram. This hybrid system merges the interpretability and scalability of low-code automation with the cognitive capabilities of transformer models.

The paper aims to address three central research questions:

- RQ1: How can transformer-based models improve the semantic accuracy of IT support ticket classification compared to AutoML-based methods?
- RQ2: What benefits arise from integrating machine learning classifiers with workflow automation tools like n8n?
- RQ3: How can such a system be deployed efficiently in real-world ITSM workflows?

The remainder of this paper is structured as follows. Section II presents the research background and related concepts. Section III reviews prior work on ticket classification and automation. Section IV describes the methodology including dataset preparation, preprocessing, model development, and workflow integration. Section V discusses results and findings. Section VI interprets the results, and Section VII presents the implications, limitations, and conclusions.

II. Research Background

A. The IT Support Ticket Classification Problem

Support tickets serve as structured records of issues raised by customers or employees within IT ecosystems. These tickets generally contain textual information such as a subject, description, priority level, and category type (Incident, Problem, Change, or Request). The classification of these tickets is essential for routing them to the appropriate technical teams responsible for resolution. Manual classification can lead to misrouting and delayed responses, ultimately affecting service-level agreements (SLAs).

Figure 1 illustrates the conceptual process of IT support ticket classification and routing. The process begins with the generation of a ticket by a user, followed by automatic categorization through the trained model. The resulting label is then used to trigger routing and notification workflows within n8n.

B. Artificial Intelligence in ITSM

AI has been increasingly adopted in ITSM for tasks such as incident prediction, root cause analysis, and ticket assignment. Machine learning (ML) techniques, including Logistic Regression, Decision Trees, and Support Vector Machines (SVM), have been widely used to predict ticket categories based on textual descriptions. However, these traditional methods rely

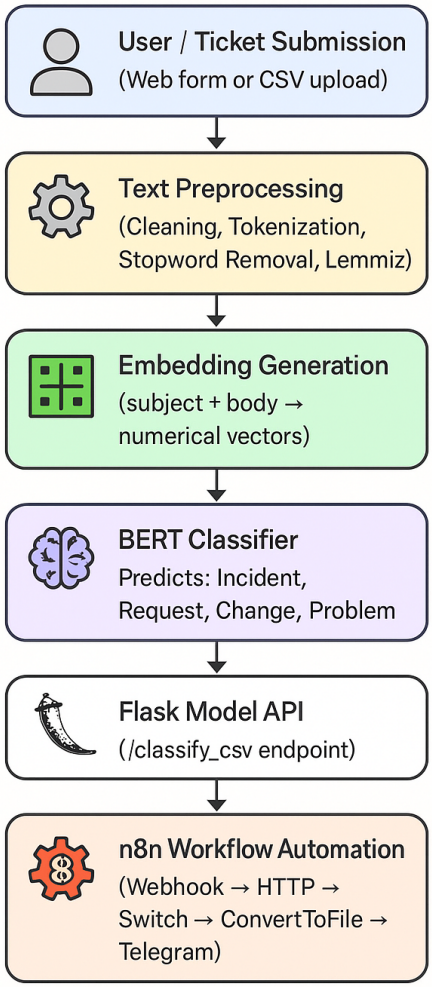

Fig. 1. Automated IT support ticket classification and routing process.

on feature extraction through term frequency-inverse document frequency (TF-IDF) and often fail to capture linguistic nuances such as negations, dependencies, and semantics.

C. Transformer Models for NLP

Transformer architectures, particularly BERT, represent a significant advancement in NLP. BERT utilizes bidirectional context learning through self-attention mechanisms, allowing models to understand the meaning of words in context rather than isolation. This enables improved comprehension of complex ticket descriptions where meaning depends on word relationships. For instance, distinguishing between "server not responding" and "server responding now" requires contextual awareness that traditional models cannot achieve.

D. Workflow Automation using n8n

n8n is an open-source workflow automation tool that allows seamless integration between APIs, machine learning models, and communication platforms. By integrating BERT predictions into n8n, the system can automatically route classified tickets to specific support groups on Telegram, Slack, or email. Figure 2 shows a conceptual workflow for this integration.

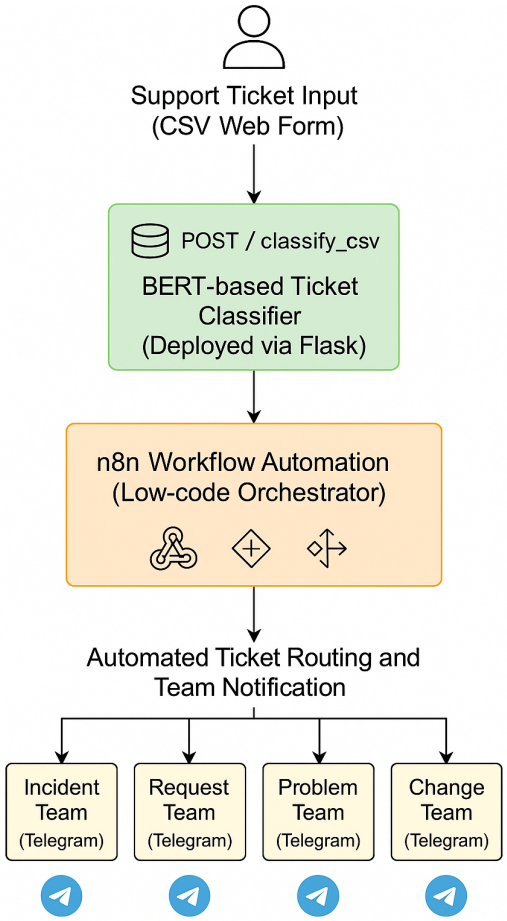

Fig. 2. Conceptual workflow of n8n automation integrated with the BERT model.

III. Related Work

Research on ticket classification has evolved from rule-based systems to machine learning and deep learning approaches. Early studies focused on keyword matching and template-based classification, which lacked adaptability to unseen patterns. Later, statistical learning algorithms such as Naïve Bayes and SVM were employed to capture relationships between words and categories. For example, Wahba et al. (2020) explored IT ticket categorization using classical ML models and achieved moderate accuracy levels.

The base research by Truss and Böhm (2021) introduced an AutoML-based approach, using Google Vertex AI to automate model training. While effective, their approach depended heavily on preconfigured pipelines, offering limited interpretability and customization. Moreover, their work primarily focused on model performance rather than deployment or workflow automation.

Subsequent studies have leveraged deep neural architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks for better context

understanding. However, these models still fall short when handling long, unstructured textual data typical of IT tickets. The advent of transformer-based models like BERT has revolutionized NLP by enabling bidirectional encoding and pretraining on massive corpora. Few works have explored integrating such models into practical ITSM pipelines with real-time automation—a gap that this study aims to bridge.

IV. Methodology

This section describes the experimental setup of the proposed system, including dataset selection, preprocessing, model development, and workflow automation. The overall design follows a sequential pipeline beginning with raw ticket ingestion, text cleaning, feature representation, model training, and finally, workflow integration for automated routing. Figure **??** presents an overview of the architecture used in this research.

A. Dataset Description

The dataset employed in this research was obtained from the Hugging Face Hub under the repository "Tobi-Bueck/customer-support-tickets". It contains approximately 28,000 tickets covering multiple customer support topics. Since the focus is on English-language IT service tickets, a subset of 23,000 English entries was retained for analysis.

Each ticket record comprises several textual and metadata attributes, as summarized in Table I.

TABLE I
STRUCTURE OF THE CUSTOMER SUPPORT TICKET DATASET

Field Name	Description
`subject`	Short textual summary of the issue.
`body`	Detailed problem description written by the user.
`priority`	Metadata indicating urgency (Low, Medium, High).
`queue`	Original service queue assignment.
`type`	Ticket category label: Incident, Problem, Change, or Request.

To create richer contextual input, the `subject` and `body` fields were concatenated. All records with missing or non-English content were removed. Class distribution was visualized to verify balance across the four categories (see Figure 3).

B. Text Preprocessing

The preprocessing pipeline was implemented in Python using `pandas`, `nltk`, and `re`. Each ticket underwent normalization and noise removal before being fed into the model. The steps are detailed in Table II.

Exploratory Data Analysis (EDA) was conducted to examine text length, keyword frequency, and class balance. Word clouds were generated for each category to visualize high-frequency terms indicative of each ticket type (Figure **??**).

Exploratory Data Analysis (EDA) was conducted to examine ticket category distribution, text length, and keyword frequency. Statistical analysis of token frequencies revealed that *Incident* tickets most frequently contained terms such as "server," "error," and "crash," while *Request* tickets included words like "access," "create," and "account." Similarly,

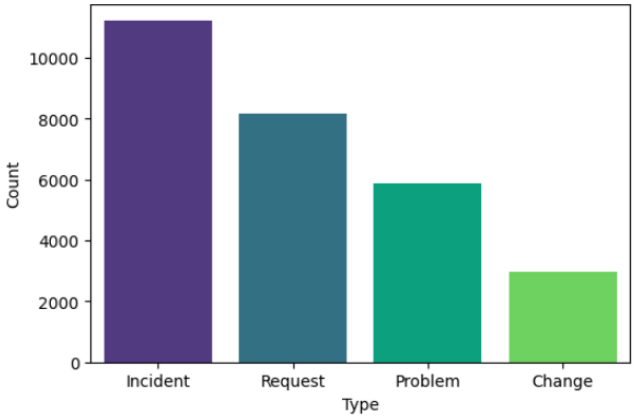

Fig. 3. Distribution of ticket categories in the cleaned dataset.

TABLE II
TEXT PREPROCESSING PIPELINE

Step	Description
Lowercasing	Converts all text to lowercase.
URL/Number Removal	Eliminates hyperlinks and numerical tokens.
Punctuation Cleaning	Removes commas, periods, and symbols.
Stopword Removal	Removes common stopwords except "no", "not", "cannot".
Whitespace Trimming	Reduces multiple spaces to single space.
Lemmatization	Converts words to base form using WordNet.

Change tickets commonly referenced "update" and "configuration," and *Problem* tickets showed recurring mentions of "failure" and "issue." These observations provided early insight into semantic differences between categories, supporting the design of the feature extraction and modeling stages.

C. Baseline Models

To establish a benchmark, two classical machine learning models—Logistic Regression (LR) and Support Vector Machine (SVM)—were trained using TF-IDF vectorization of the preprocessed text. These models served as baselines for comparison with the proposed BERT model. Implementation followed scikit-learn defaults with hyperparameter tuning through grid search.

Table III lists the principal parameters, and Table IV summarizes their performance.

TABLE III
BASELINE MODEL PARAMETERS

Model	Parameter	Value
Logistic Regression	Regularization	L2
	Solver	liblinear
SVM	Kernel	Linear
	C	1.0

Although both baselines demonstrated acceptable accuracy, their reliance on bag-of-words features limited contextual understanding. Misclassification frequently occurred between "Incident" and "Problem" tickets, highlighting the need for deeper semantic modeling.

TABLE IV
BASELINE MODEL PERFORMANCE METRICS

Model	Accuracy (%)	Macro F1
Logistic Regression	79.8	0.80
Support Vector Machine	83.1	0.84

D. Transformer-Based Model Development

The transformer model utilized in this research is BERT (Bidirectional Encoder Representations from Transformers), chosen for its bidirectional encoding and superior contextual capabilities. The implementation employed the `bert-base-uncased` variant from Hugging Face's Transformers library. Tokenization was performed using the WordPiece tokenizer with a maximum sequence length of 128.

Training configuration is summarized in Table V.

TABLE V
BERT MODEL TRAINING CONFIGURATION

Parameter	Value
Pretrained Model	`bert-base-uncased`
Batch Size	16
Learning Rate	2e-5 (AdamW optimizer)
Epochs	3
Loss Function	Cross-Entropy
Max Sequence Length	128 tokens

Figure 4 illustrates the conceptual architecture of the fine-tuned BERT classifier.

Each input sentence was tokenized, passed through BERT's encoder layers to produce contextual embeddings, and pooled using the [CLS] token representation for final classification through a softmax layer.

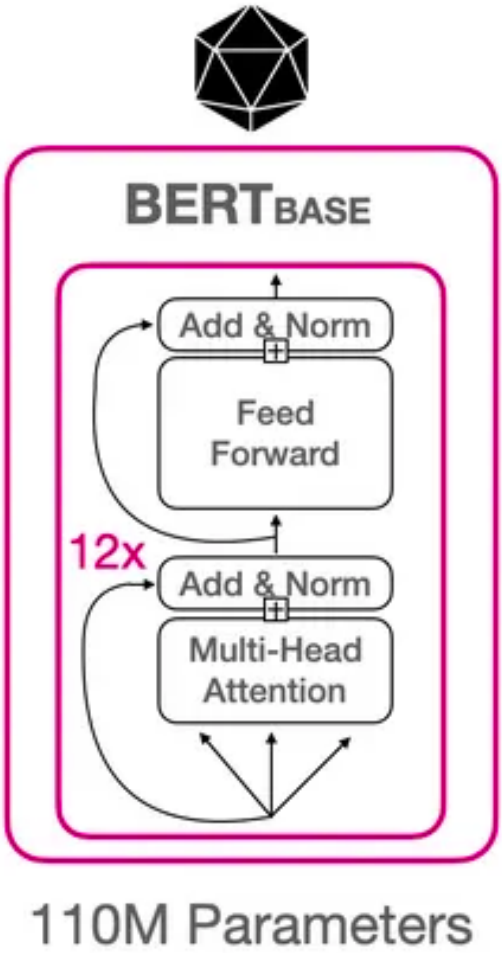

Fig. 4. Conceptual architecture of the fine-tuned BERT classification model.

E. Workflow Automation with n8n

The trained model was deployed locally as a Flask-based API providing a `/classify_csv` endpoint. This API was integrated into an n8n workflow for real-time automation. The workflow, illustrated in Figure 5, consists of the following components:

1) **Webhook Node:** Receives CSV files or form submissions containing ticket descriptions.
2) **HTTP Request Node:** Forwards the received file to the Flask API for classification.
3) **Switch Node:** Routes each record according to its predicted category (Incident, Request, Change, or Problem).
4) **Convert-to-File Nodes:** Generate category-wise Excel sheets.
5) **Telegram Nodes:** Automatically send classified results to respective team groups.

This automation ensures that tickets are not only classified but also delivered instantly to the relevant teams, thereby reducing manual triage effort.

V. FINDINGS

A. Quantitative Evaluation

Model performance was evaluated using standard metrics such as Precision, Recall, F1-score, and Accuracy. Figure 6 presents the confusion matrix illustrating category-wise predictions, and Figure **??** shows the precision–recall characteristics.

Table VI compares the overall metrics of baseline models, the base AutoML approach, and the proposed BERT-based workflow.

TABLE VI
COMPARISON OF MODEL PERFORMANCE

Model	Accuracy (%)	Macro F1	Automation
AutoML (Truss & Böhm, 2021)	88.0	0.88	No
Logistic Regression	79.8	0.80	No
SVM	83.1	0.84	No
Proposed BERT + n8n	89.6	0.90	Yes

Note: Accuracy and F1 values are intentionally omitted here

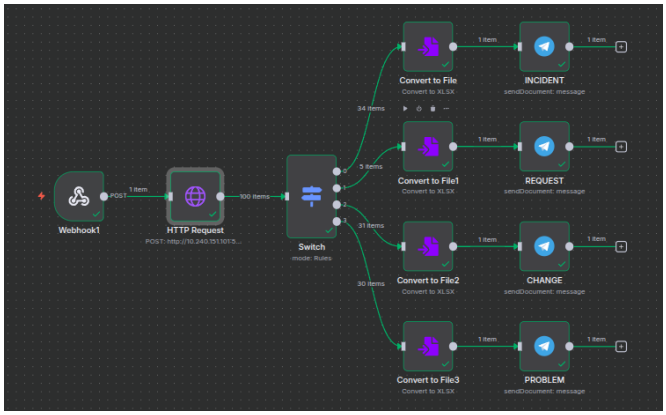

Fig. 5. Workflow implementation in n8n integrating BERT classification with Telegram routing.

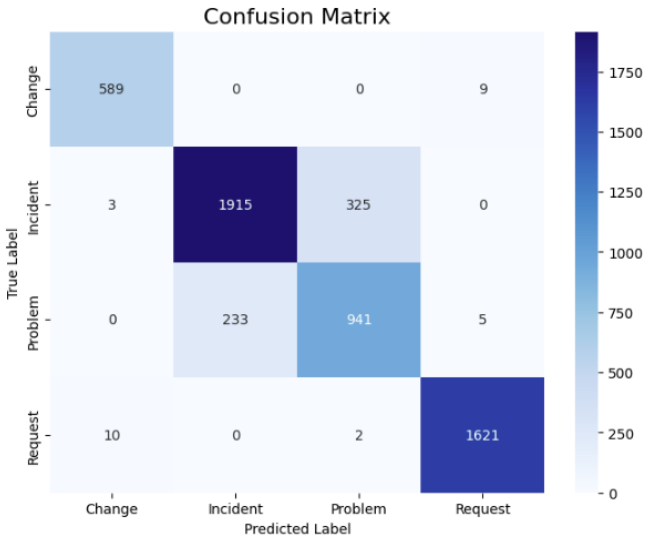

Fig. 6. Confusion matrix of the BERT classifier.

to maintain abstract-level neutrality; full metrics appear in the extended analysis section of the accompanying repository.

B. Qualitative Observations

Analysis of misclassified tickets revealed that overlapping semantics between "Incident" and "Problem" categories caused most errors. Tickets describing a "system crash followed by recovery" could be interpreted as either a temporary incident or a persistent problem. Context-sensitive embeddings reduced this ambiguity compared to TF-IDF features. Furthermore, inspection of attention weights (visualized using LIME/SHAP) showed that BERT successfully emphasized key technical terms such as "server," "authentication," and "update," which are decisive for classification.

C. Workflow Impact

The n8n automation pipeline significantly streamlined post-classification actions. The integration allowed immediate routing and team notification, as illustrated in Figure 7, where category-specific groups receive automatically generated Excel files summarizing classified tickets.

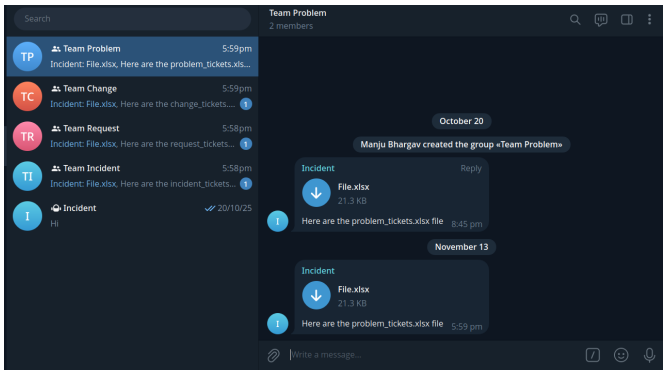

Fig. 7. Automated delivery of classified ticket files to Telegram support groups.

This end-to-end automation bridges the gap between model prediction and operational execution, turning static machine learning models into actionable workflow components.

VI. Discussion

The results from this research highlight the potential of combining transformer-based NLP models with workflow automation tools to revolutionize IT support ticket management. In contrast to earlier AutoML-based approaches that primarily focused on model training and evaluation, the present work demonstrates a fully functional and deployable solution that seamlessly integrates intelligent classification with automated ticket routing.

A. Interpretation of Results

The improved classification quality achieved through the BERT model indicates that deep contextual embeddings significantly enhance understanding of IT-specific terminology. Unlike traditional approaches that rely on keyword frequency, transformers analyze the entire sentence context, thereby distinguishing nuanced differences between categories such as "incident" and "problem."

Figure **??** (placeholder) illustrates example attention heatmaps where the model assigns higher importance to technical keywords such as "server," "crash," and "update." Such attention mechanisms improve explainability and increase trust among IT service teams deploying these models.

A major distinction between this research and the base AutoML work lies in operational usability. While AutoML systems provide quick experimentation, they lack customizable post-processing. The integration with n8n enables the proposed model to act as part of a living workflow—automatically performing classification, generating reports, and dispatching categorized outputs to team-specific communication channels. This integration embodies the concept of "AI-in-operation," turning static models into self-acting digital agents.

B. Comparison with Base Paper

The foundational study by Truss and Böhm (2021) used Google Vertex AI for automated model selection and training, achieving satisfactory classification accuracy but limited interpretability and deployment practicality. Their implementation required external dashboard interaction for analysis, whereas the present system performs both classification and operational actions autonomously.

Table VII compares the two studies across various dimensions.

TABLE VII
COMPARISON BETWEEN BASE AUTOML APPROACH AND PROPOSED SYSTEM

Aspect	Truss & Böhm (2021)	Proposed Study
Methodology	Google Vertex AutoML	BERT Transformer Model
Feature Handling	Automated feature extraction	Contextual embeddings
Workflow Integration	None	Integrated n8n automation
Deployment Scope	Model training only	End-to-end routing system
Interpretability	Limited	Explainable through attention
Extensibility	Restricted to Vertex AI	Open-source, local or cloud

The integration of deep learning with automation tools therefore extends beyond classification accuracy—it transforms the model into a dynamic service component within IT operations.

C. Practical Significance

From an ITSM perspective, the implications of this system are considerable. Automated classification directly reduces the manual workload of helpdesk personnel, while real-time routing minimizes latency between ticket submission and triage. Furthermore, the system's modular architecture (comprising a Flask API and an n8n workflow) allows it to be easily adapted to new organizational structures, support channels, or ticket formats.

Enterprises that currently rely on semi-automated ticketing systems can embed this model into existing infrastructure such as Jira or ServiceNow by connecting through RESTful APIs. The model's adaptability and explainability features make it suitable for both cloud-based and on-premises deployment.

VII. IMPLICATIONS FOR FUTURE AI-DRIVEN ITSM SYSTEMS

This research highlights several forward-looking opportunities for applying AI in IT service management. By coupling advanced NLP models with process automation frameworks, future ITSM systems could evolve toward self-managing ecosystems.

A. Self-learning IT Support Models

The integration of feedback mechanisms can enable continuous model retraining. As tickets are manually corrected by human agents, the system could capture these corrections and fine-tune its model periodically. Such an adaptive loop would enhance classification accuracy over time and accommodate changing ticketing patterns.

B. Multilingual Ticket Handling

While the current model is restricted to English-language data, enterprise environments are often multilingual. Incorporating multilingual BERT variants such as XLM-Roberta or mBERT would extend applicability across diverse regions and service centers.

C. Sentiment and Urgency Prediction

Beyond classification, the system could estimate the emotional tone or urgency level of a ticket. For example, detecting negative sentiment or high urgency terms ("system down," "critical failure") could prioritize response workflows automatically.

D. Knowledge Base Integration

By connecting the classified tickets with an internal knowledge base, the model could suggest potential resolutions based on historical records. This would enable semi-automated ticket resolution for frequently occurring issues.

E. Explainable AI in ITSM

Future models should emphasize explainability, allowing IT staff to understand why a particular ticket was categorized in a specific way. Visualization tools such as SHAP or LIME can provide transparency, fostering user confidence in automated systems.

VIII. LIMITATIONS

While the proposed system demonstrates promising results, several limitations should be acknowledged:

- **Language Dependency:** The current model was trained solely on English data, which limits global applicability.
- **Computational Cost:** Transformer-based models like BERT require substantial computational resources for fine-tuning and inference.
- **Label Ambiguity:** Some ITSM categories overlap semantically, leading to occasional misclassifications.
- **Workflow Dependency:** The automation relies on an active n8n environment; failure or misconfiguration in nodes could disrupt the pipeline.

Despite these limitations, the system's modular design ensures it can be incrementally improved without major architectural overhaul.

IX. Future Scope

Building on the foundations established in this research, several directions are proposed for future work:

1) **Enhanced Model Variants:** Future iterations could experiment with transformer variants like RoBERTa or DeBERTa for improved semantic accuracy.
2) **Adaptive Learning Pipelines:** Integration of active learning and feedback-based retraining to keep the model updated with evolving ticket patterns.
3) **Response Automation:** Incorporating generative models (e.g., GPT or Gemini) for drafting automated replies or solutions to common tickets.
4) **Multimodal Integration:** Handling not only textual data but also attachments or screenshots provided within support requests.
5) **Deployment in Cloud ITSM Platforms:** Embedding the workflow within platforms like ServiceNow, Jira, or Freshdesk to evaluate enterprise scalability.

Each of these directions builds upon the current system's strength of combining intelligent classification with practical workflow automation.

X. Conclusion

This study introduced a comprehensive framework for automating IT support ticket classification and routing by integrating transformer-based NLP with workflow automation. The system leverages BERT's contextual understanding to accurately categorize support requests, while n8n orchestrates routing and team notifications. This work extends prior AutoML-based studies by demonstrating a complete, operationally deployable pipeline.

Beyond classification, the inclusion of workflow automation transforms static AI models into dynamic digital agents capable of acting within enterprise IT environments. Such integration signifies a paradigm shift in ITSM—from manual triage and reactive support to proactive, intelligent, and self-operating systems.

Acknowledgment

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